

Модуль 4: Docker Storage (Volumes) и Networking

Задача (Теория): Почему данные пропадают?

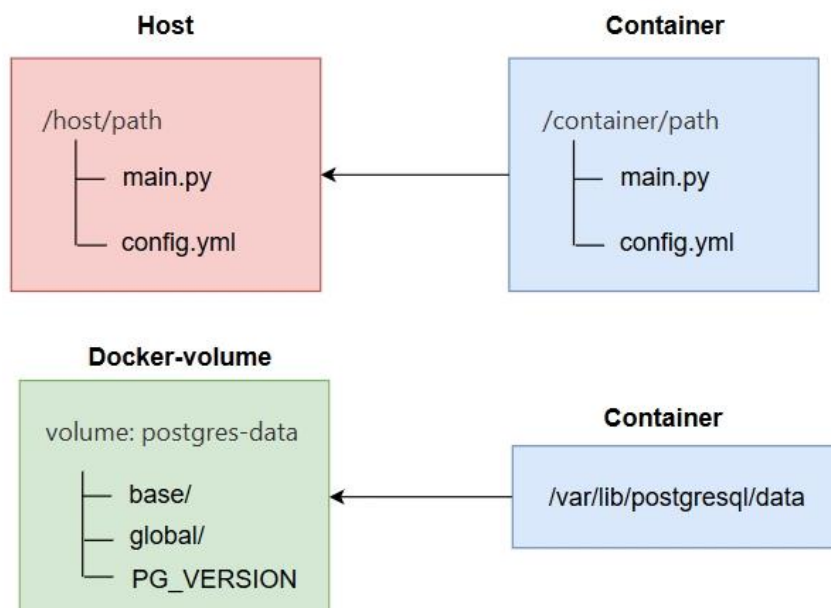
Задание: `docker run -it ubuntu bash`, создать `touch /data/test.txt`, `exit`. `docker rm [container]`. `docker run ...` (тот же образ) — файла `/data/test.txt` нет.

```
root@Cetus:/opt/basic# docker run -dit ubuntu bash
822abf789593aaf162748df5e46774297d5a64da4bf054ad56b5fef6daab4db
root@Cetus:/opt/basic# docker exec -it 822abf789593 bash
root@822abf789593:/# mkdir /data
root@822abf789593:/# touch /data/test.txt
root@822abf789593:/# ls /data
test.txt
root@822abf789593:/# exit
exit
```

```
root@Cetus:/opt/basic# docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS
822abf789593   ubuntu        "bash"                  About a minute ago    Up About a minute
16eef8023d28   my-dotnet-app "dotnet my-app.dll"     2 hours ago       Up 2 hours
1e90e20b3c1d   my-app:1.1    "/docker-entrypoint. ..." 2 hours ago       Up 2 hours
8bc92e53a96c   my-app:1.2    "/docker-entrypoint. ..." 2 hours ago       Up 2 hours (healthy)
root@Cetus:/opt/basic# docker stop amazing_carver
amazing_carver
root@Cetus:/opt/basic# docker rm amazing_carver
amazing_carver
root@Cetus:/opt/basic# docker run -it ubuntu bash
root@8e27d62a840f:/# ls /data
ls: cannot access '/data': No such file or directory
```

Задача (Теория): Bind Mount vs. Volume.

Задание: Нарисовать схему, объясняющую разницу.



Задача (Bind Mount): "Горячая" перезагрузка (Hot Reload).

Задание: Запустить Python/Node приложение, используя Bind Mount: `docker run -p 5000:5000 -v $(pwd):/app my-app:1.0` (может требоваться `uvicorn --reload` или `nodemon`).

```
root@Cetus:/opt/basic/module4# docker run -d -p 5000:5000 -v $(pwd):/app my-app:1.3
59739635dffc29127e83fdedf054852a8eed4f15e48b9d8b145af6cc66c7c6ff
root@Cetus:/opt/basic/module4# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
59739635dffc   my-app:1.3     "python app.py"         3 seconds ago Up 2 seconds  0.0.0.0:5000->5000/tcp
1e90e20b3c1d   my-app:1.1     "/docker-entrypoint.    3 hours ago   Up 3 hours    80/tcp
8bc92e53a96c   my-app:1.2     "/docker-entrypoint.    3 hours ago   Up 3 hours (healthy) 80/tcp
root@Cetus:/opt/basic/module4# nano test.txt
root@Cetus:/opt/basic/module4# docker exec -it dreamy_tesla bash
root@59739635dffc:/app# ls
Dockerfile app.py requirements.txt test.txt
root@59739635dffc:/app# cat test.txt
Rewrite me
!!!!!!!!!!!!!!!!!!!!!!
root@59739635dffc:/app# exit
exit
root@Cetus:/opt/basic/module4# nano test.txt
root@Cetus:/opt/basic/module4# root@Cetus:/opt/basic/module4# docker exec -it dreamy_tesla bash
root@59739635dffc:/app# cat test.txt
Rewrite me
root@59739635dffc:/app# exit
exit
```

Задача (Volume): Создать "именованный" Volume: `docker volume create postgres-data`.

```
root@Cetus:/opt/basic/module4# docker volume create postgres-data
postgres-data
root@Cetus:/opt/basic/module4# docker volume ls
DRIVER      VOLUME NAME
local       postgres-data
local       test_pgdata
root@Cetus:/opt/basic/module4#
```

Задача (Volume): Запустить PostgreSQL с Volume.

Задание: `docker run -d -p 5432:5432 -v postgres-data:/var/lib/postgresql/data -e POSTGRES_PASSWORD=mysecret postgres`.

```
root@Cetus:/opt/basic/module4# docker run -d --name pg-test -e POSTGRES_PASSWORD=mysecret -v postgres-data:/var/lib/postgresql/data postgres:17
Unable to find image 'postgres:17' locally
17: Pulling from library/postgres
119d43eef815: Already exists
88605c048c10: Pull complete
aa732be6e8c4: Pull complete
2967a6781cea: Pull complete
1aec77daa624: Pull complete
1bf67474c634: Pull complete
8721b06a7eca: Pull complete
4e996ff323f4: Pull complete
49a9373ae96f: Pull complete
90cc988fa305: Pull complete
102f8f91db14: Pull complete
2f656de9a048: Pull complete
8fe4bbb3d2a7: Pull complete
4cbb65d57930: Pull complete
Digest: sha256:7352e0c4d62bbac8aa69d95e40220a60967c4a19f9c4f65b4d118175f7ce9e3b
Status: Downloaded newer image for postgres:17
467c3ebecab07002005b0578a3118e70bc2f97ebd8cef081761e3f49371dd64
root@Cetus:/opt/basic/module4# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
467c3ebecab0   postgres:17    "docker-entrypoint.s..." 11 seconds ago Up 11 seconds  5432/tcp     pg-test
59739635dffc   my-app:1.3     "python app.py"         12 minutes ago Up 12 minutes  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp  dreamy_t
1e90e20b3c1d   my-app:1.1     "/docker-entrypoint.    3 hours ago   Up 3 hours    80/tcp, 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp  my-app-1
8bc92e53a96c   my-app:1.2     "/docker-entrypoint.    3 hours ago   Up 3 hours (healthy) 80/tcp, 0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp  my-app-2
```

```

root@Cetus:/opt/basic/module4# docker logs pg-test
The files belonging to this database system will be owned by user "postgres".
This user must also own the server process.

The database cluster will be initialized with locale "en_US.utf8".
The default database encoding has accordingly been set to "UTF8".
The default text search configuration will be set to "english".

Data page checksums are disabled.

fixing permissions on existing directory /var/lib/postgresql/data ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default "max_connections" ... 100
selecting default "shared_buffers" ... 128MB
selecting default time zone ... Etc/UTC
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... ok
initdb: warning: enabling "trust" authentication for local connections
initdb: hint: You can change this by editing pg_hba.conf or using the option -A, or --auth-local and --auth-host, the next time you run initdb.
syncing data to disk ... ok

Success. You can now start the database server using:

    pg_ctl -D /var/lib/postgresql/data -l logfile start

waiting for server to start...2026-01-29 20:42:04.978 UTC [48] LOG:  starting PostgreSQL 17.7 (Debian 17.7-3.pgdg13+1) on x86_64-pc-linux-gnu, compiled by gcc (Deb
2026-01-29 20:42:04.983 UTC [48] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
2026-01-29 20:42:04.997 UTC [51] LOG:  database system was shut down at 2026-01-29 20:42:03 UTC
2026-01-29 20:42:05.000 UTC [48] LOG:  database system is ready to accept connections
done
server started

/usr/local/bin/docker-entrypoint.sh: ignoring /docker-entrypoint-initdb.d/*

2026-01-29 20:42:05.168 UTC [48] LOG:  received fast shutdown request
waiting for server to shut down...2026-01-29 20:42:05.172 UTC [48] LOG:  aborting any active transactions
2026-01-29 20:42:05.174 UTC [48] LOG:  background worker "logical replication launcher" (PID 54) exited with exit code 1
2026-01-29 20:42:05.177 UTC [49] LOG:  shutting down
2026-01-29 20:42:05.182 UTC [49] LOG:  checkpoint starting: shutdown immediate
2026-01-29 20:42:05.207 UTC [49] LOG:  checkpoint complete: wrote 3 buffers (0.0%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.007 s, sync=0.003 s, total=0
2026-01-29 20:42:05.215 UTC [48] LOG:  database system is shut down
done
server stopped

PostgreSQL init process complete; ready for start up.

2026-01-29 20:42:05.309 UTC [1] LOG:  starting PostgreSQL 17.7 (Debian 17.7-3.pgdg13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
2026-01-29 20:42:05.312 UTC [1] LOG:  listening on IPv4 address "0.0.0.0", port 5432
2026-01-29 20:42:05.312 UTC [1] LOG:  listening on IPv6 address ":::", port 5432
2026-01-29 20:42:05.321 UTC [1] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
2026-01-29 20:42:05.332 UTC [62] LOG:  database system was shut down at 2026-01-29 20:42:05 UTC
2026-01-29 20:42:05.341 UTC [1] LOG:  database system is ready to accept connections
root@Cetus:/opt/basic/module4# docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
867c3ebecab0	postgres:17	"docker-entrypoint.s..."	48 seconds ago	Up 48 seconds	5432/tcp	pg-test
59739635dffc	my-app:1.3	"python app.py"	13 minutes ago	Up 13 minutes	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	dreamy_tesla
1e90e20b3c1d	my-app:1.1	"docker-entrypoint.s..."	3 hours ago	Up 3 hours	80/tcp, 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp	my-app-multi
8bc92e53a96c	my-app:1.2	"docker-entrypoint.s..."	3 hours ago	Up 3 hours (healthy)	80/tcp, 0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp	my-app-non-root

```

root@Cetus:/opt/basic/module4#

```

Задача (Проверка Volume): docker stop/rm контейнер Postgres. Запустить команду из Задачи 5 еще раз.

```

root@Cetus:/opt/basic/module4# docker stop pg-test
pg-test
root@Cetus:/opt/basic/module4# docker rm pg-test
pg-test
root@Cetus:/opt/basic/module4# docker run -d --name pg-test -e POSTGRES_PASSWORD=mysecret -v postgres-data:/var/lib/postgresql/data postgres:17
4c99b61b33e84169f4f87bb04f61c9341236f8f05f0578ed5614dcb4a9c3b99e
root@Cetus:/opt/basic/module4# docker logs pg-test

PostgreSQL Database directory appears to contain a database; Skipping initialization

2026-01-29 20:44:54.307 UTC [1] LOG:  starting PostgreSQL 17.7 (Debian 17.7-3.pgdg13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0
2026-01-29 20:44:54.307 UTC [1] LOG:  listening on IPv4 address "0.0.0.0", port 5432
2026-01-29 20:44:54.307 UTC [1] LOG:  listening on IPv6 address ":::", port 5432
2026-01-29 20:44:54.314 UTC [1] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
2026-01-29 20:44:54.324 UTC [29] LOG:  database system was shut down at 2026-01-29 20:44:42 UTC
2026-01-29 20:44:54.332 UTC [1] LOG:  database system is ready to accept connections
root@Cetus:/opt/basic/module4#

```


Задача (Теория Сети): Сеть bridge (по умолчанию).

Задание: Запустить 2 ubuntu контейнера (sleep 1000). docker inspect и найти их IP. Зайти в первый (docker exec -it [id] bash) и ping второй по IP.

```
root@Cetus:/opt/basic/module4# docker run -d --name u1 ubuntu sleep 1000
2 ubuntu sleep 1000
e2865776d546c447e5f3f0c1fee00f01cdc7c7ea92a73468136c48756959da95
root@Cetus:/opt/basic/module4# docker run -d --name u2 ubuntu sleep 1000
66c4f4ae9db3392e49e43f034348482f1fbdac5e5a65cd14615cf76d7db41f3b
root@Cetus:/opt/basic/module4# docker inspect u1 | grep IPAddress
p IPAddress
    "SecondaryIPAddresses": null,
    "IPAddress": "172.17.0.6",
        "IPAddress": "172.17.0.6",
root@Cetus:/opt/basic/module4# docker inspect u2 | grep IPAddress
    "SecondaryIPAddresses": null,
    "IPAddress": "172.17.0.7",
        "IPAddress": "172.17.0.7",
```

```
root@Cetus:/opt/basic/module4# docker exec -it u1 bash
root@e2865776d546:/# ping 172.17.0.7
PING 172.17.0.7 (172.17.0.7) 56(84) bytes of data.
64 bytes from 172.17.0.7: icmp_seq=1 ttl=64 time=0.132 ms
64 bytes from 172.17.0.7: icmp_seq=2 ttl=64 time=0.054 ms
64 bytes from 172.17.0.7: icmp_seq=3 ttl=64 time=0.050 ms
64 bytes from 172.17.0.7: icmp_seq=4 ttl=64 time=0.057 ms
64 bytes from 172.17.0.7: icmp_seq=5 ttl=64 time=0.061 ms
^C
--- 172.17.0.7 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4083ms
rtt min/avg/max/mdev = 0.050/0.070/0.132/0.030 ms
root@e2865776d546:/#
```

Задача (Сеть bridge): Создать свою bridge сеть: docker network create my-app-net.

```
root@Cetus:/opt/basic/module4# docker network create my-app-net
8abeb7e98b934e611d772725428dbfb5579030ed231e8d9597255cf95ba0b953
root@Cetus:/opt/basic/module4# docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
d020e6f345ef        bridge              bridge              local
ff4f89ecb320        host                host                local
8abeb7e98b93        my-app-net          bridge              local
1a5e8eeb8243        none                null                local
c9437c15fc80        test-project_default bridge              local
cabdd02daa4f        test_default        bridge              local
```

Задача (Подключение к сети): Запустить Postgres в этой сети (--network my-app-net --name pg-db).

```
root@Cetus:/opt/basic/module4# docker run -d --network my-app-net --name pg-db -e POSTGRES_PASSWORD=mysecret postgres
cb6ee7ae9e0005bf75dd5bae88d2f1c2fd3855f28b8e47f074a8544c18724bdd
root@Cetus:/opt/basic/module4# docker network inspect my-app-net
[
  {
    "Name": "my-app-net",
    "Id": "8abeb7e98b934e611d772725428dbfb5579030ed231e8d9597255cf95ba0b953",
    "Created": "2026-01-29T20:51:00.4573391Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.20.0.0/16",
          "Gateway": "172.20.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {
      "cb6ee7ae9e0005bf75dd5bae88d2f1c2fd3855f28b8e47f074a8544c18724bdd": {
        "Name": "pg-db",
        "EndpointID": "9d83b4daf3c5ddc86a36e72ed05217648502410a924cd250eefcc26d6275d5ac",
        "MacAddress": "82:d0:05:ac:1a:bf",
        "IPv4Address": "172.20.0.2/16",
        "IPv6Address": ""
      }
    },
    "Options": {},
    "Labels": {}
  }
]
```


Задача (DNS-Discovery): Запустить ubuntu в этой же сети (--network my-app-net). apt update && apt install dnsutils -y. Выполнить ping pg-db.

```
root@Cetus:/opt/basic/module4# docker run -it --network my-app-net ubuntu bash
root@54ba98d58ac5:/# apt update
Get:1 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [34.8 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1194 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3219 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [3008 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2171 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1973 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [38.1 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [49.5 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [34.6 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1776 kB]
Fetched 35.7 MB in 4s (8290 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
1 package can be upgraded. Run 'apt list --upgradable' to see it.
root@54ba98d58ac5:/# apt install iputils-ping -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcap2-bin libpam-cap
The following NEW packages will be installed:
  iputils-ping libcap2-bin libpam-cap
0 upgraded, 3 newly installed, 0 to remove and 1 not upgraded.
Need to get 91.2 kB of archives.
After this operation, 322 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcap2-bin amd64 1:2.66-5ubuntu2.2 [34.2 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 iputils-ping amd64 3:20240117-1ubuntu0.1 [44.6 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libpam-cap amd64 1:2.66-5ubuntu2.2 [12.5 kB]
Fetched 91.2 kB in 1s (119 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libcap2-bin.
(Reading database ... 4381 files and directories currently installed.)
Preparing to unpack .../libcap2-bin_1%3a2.66-5ubuntu2.2_amd64.deb ...
Unpacking libcap2-bin (1:2.66-5ubuntu2.2) ...
Selecting previously unselected package iputils-ping.
Preparing to unpack .../iputils-ping_3%3a20240117-1ubuntu0.1_amd64.deb ...
Unpacking iputils-ping (3:20240117-1ubuntu0.1) ...
Selecting previously unselected package libpam-cap:amd64.
Preparing to unpack .../libpam-cap_1%3a2.66-5ubuntu2.2_amd64.deb ...
Unpacking libpam-cap:amd64 (1:2.66-5ubuntu2.2) ...
Setting up libcap2-bin (1:2.66-5ubuntu2.2) ...
Setting up libpam-cap:amd64 (1:2.66-5ubuntu2.2) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 8.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl/5.38 /usr/share/perl/5.38 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8.)
debconf: falling back to frontend: Teletype
Setting up iputils-ping (3:20240117-1ubuntu0.1) ...
```

```
root@54ba98d58ac5:/# ping pg-db
PING pg-db (172.20.0.2) 56(84) bytes of data.
64 bytes from pg-db.my-app-net (172.20.0.2): icmp_seq=1 ttl=64 time=0.129 ms
64 bytes from pg-db.my-app-net (172.20.0.2): icmp_seq=2 ttl=64 time=0.050 ms
64 bytes from pg-db.my-app-net (172.20.0.2): icmp_seq=3 ttl=64 time=0.082 ms
64 bytes from pg-db.my-app-net (172.20.0.2): icmp_seq=4 ttl=64 time=0.070 ms
^C
--- pg-db ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3051ms
rtt min/avg/max/mdev = 0.050/0.082/0.129/0.029 ms
```